TLF R-Shiny Dashboard with Teal

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1 Summary of the teal R-Shiny Dashboard

This application uses the `teal` framework (specifically `teal.modules.clinical`) to create an interactive Shiny app for exploring clinical trial data. It is designed for analysts and reviewers to dynamically filter and visualize ADaM datasets.

1.1 Key Components

Component	Description
Data	Three CDISC ADaM datasets are loaded from `teal.modules.clinical`: ADSL (subject-level), ADAE (adverse events), and ADTTE (time-to-event).
Modules	Three interactive modules are included: a demographic summary table, an adverse events table, and a Kaplan-Meier plot.
Filtering	Pre-defined and interactive filters allow users to subset data globally or per module.
Header & Footer	Custom header and footer with attribution.

1.2 Modules in Detail

1. **Demographic Table (`tm_t_summary`)**
   - Data: `ADSL`
   - Arm variable: `ARM` or `ARMCD` (default `ARM`)
   - Summarized variables: `SEX`, `RACE`, `BMRKR2`, `EOSDY`, `DCSREAS`, `AGE` (default `SEX`, `RACE`)
2. **AE Table (`tm_t_events`)**
   - Data: `ADAE`
   - Arm variable: `ARM` or `ARMCD`
   - High-level term (HLT): `AEBODSYS` or `AESOC` (default `AEBODSYS`)
   - Low-level term (LLT): `AETERM` or `AEDECOD` (default `AEDECOD`)
3. **Kaplan-Meier Plot (`tm_g_km`)**
   - Data: `ADTTE` (time-to-event)
   - Arm variable: `ARM`, `ARMCD`, or `ACTARMCD` (default `ARM`)
   - Parameter: `OS` (overall survival) from `PARAMCD`

- Strata and facet: Optional grouping by `SEX` or `BMRKR2` (none selected by default)
- Plot height: adjustable (default 600px, min 400px, max 5000px)

1.3 Pre-defined Filters (`teal_slices`)

Filter	Dataset	Expression	Purpose
`saffl`	ADSL	`SAFFL == "Y"`	Safety population (fixed, anchored)
`AESER`	ADAE	`AESER` (any value)	Serious adverse events filter (anchored)
`aerel`	ADAE	`AEREL == "Y" & AETOXGR %in% c("3","4","5")`	Grade 3+ related events
`sex`	ADSL	`SEX` (fixed)	Gender filter applied globally
`age`	ADSL	`AGE >= 18 & AGE <= 30`	Young adult filter (global)

Module-specific mapping: - Demographic Table uses `saffl` - AE Table uses `saffl` and `aerel` - Global filters (`sex`, `age`) apply to all modules

1.4 Output

The app launches with `shinyApp(app$ui, app$server)`, providing a full interactive dashboard for clinical trial data exploration.

2 Complete R Code

```
# Load required package
library(teal.modules.clinical)

ADSL <- teal.modules.clinical::tmc_ex_adsl
ADAE <- teal.modules.clinical::tmc_ex_adae
ADTTE <- teal.modules.clinical::tmc_ex_adtte

app <- init(
  data = cdisc_data(
    ADSL = ADSL, ADAE = ADAE, ADTTE = ADTTE,
    code = "
      ADSL <- teal.modules.clinical::tmc_ex_adsl
      ADAE <- teal.modules.clinical::tmc_ex_adae
      ADTTE <- teal.modules.clinical::tmc_ex_adtte
```

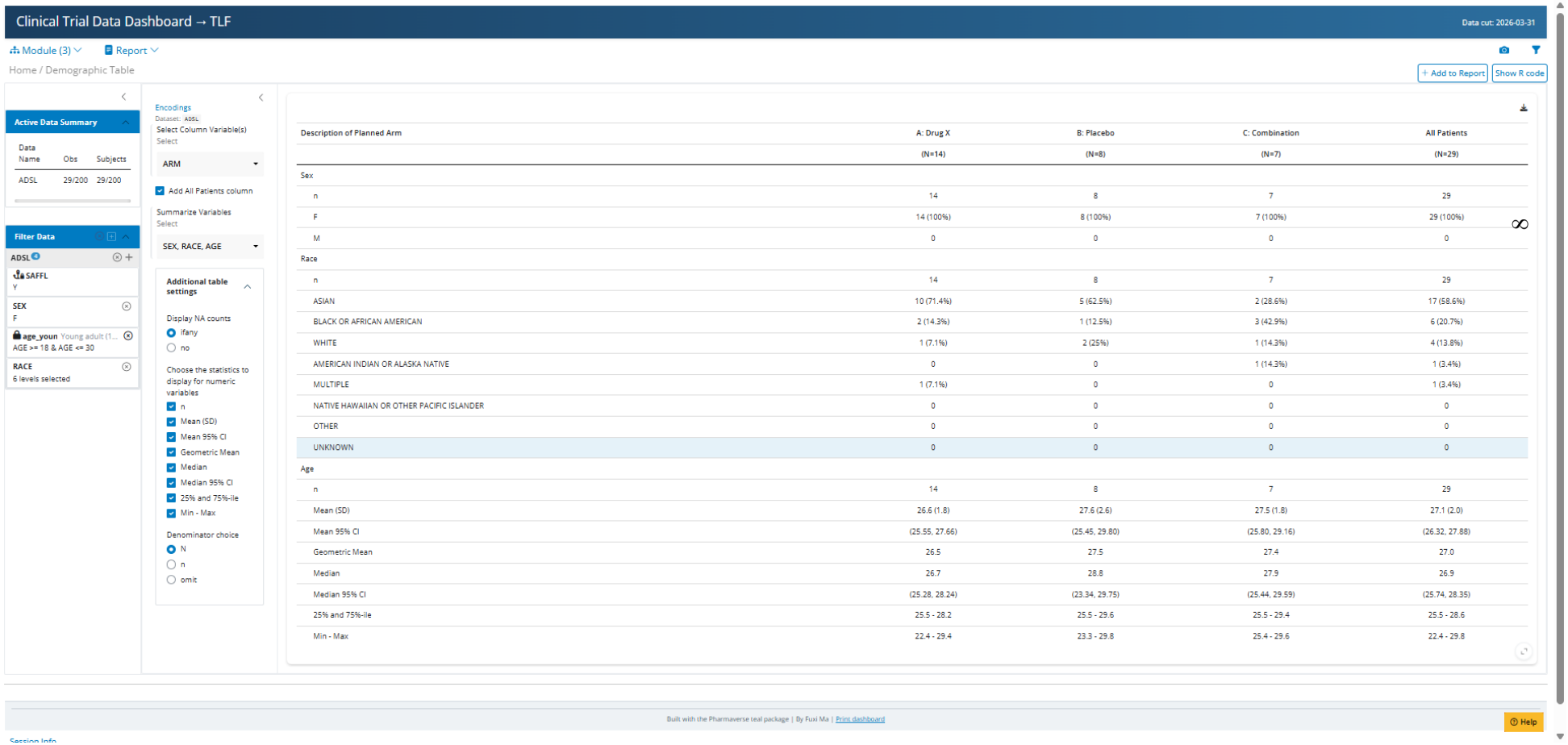
```
    "
  ),
  modules = modules(
    tm_t_summary(
      label = "Demographic Table",
      dataname = "ADSL",
      arm_var = choices_selected(
        choices = c("ARM", "ARMCD"), selected = "ARM"
      ),
      summarize_vars = choices_selected(
        choices = c("SEX", "RACE", "BMRKR2", "EOSDY", "DCSREAS", "AGE"),
        selected = c("SEX", "RACE")
      )
    ),
    tm_t_events(
      label = "AE Table",
      dataname = "ADAE",
      arm_var = choices_selected(
        choices = c("ARM", "ARMCD"), selected = "ARM"
      ),
      hlt = choices_selected(
        choices = variable_choices(ADAE, c("AEBODSYS", "AESOC")),
        selected = "AEBODSYS"
      ),
      llt = choices_selected(
        choices = variable_choices(ADAE, c("AETERM", "AEDECOD")),
        selected = "AEDECOD"
      )
    ),
    tm_g_km(
      label = "KM Plot",
      dataname = "ADTTE",
      arm_var = choices_selected(
        choices = c("ARM", "ARMCD", "ACTARMCD"),
        selected = "ARM"
      ),
      paramcd = choices_selected(
        choices = value_choices(ADTTE, "PARAMCD", "PARAM"),
        selected = "OS"
      ),
      strata_var = choices_selected(
        variable_choices(ADSL, c("SEX", "BMRKR2")), NULL
```

```r
      ),
      facet_var = choices_selected(
        variable_choices(ADSL, c("SEX", "BMRKR2")), NULL
      ),
      plot_height = c(600L, 400L, 5000L)
    )
  ),
  header = tags$h1("Clinical Trial R-Shiny App"),
  footer = tags$div(
    style = "text-align: center; margin-top: 20px;
             padding: 10px; background-color: #f5f5f5;
             font-size: 12px;",
    "Fuxi Ma", "Built with Pharmaverse teal package"
  ),
  filter = teal_slices(
    teal_slice("ADSL", "SAFFL", id = "saffl",
               selected = "Y", fixed = TRUE, anchored = TRUE),
    teal_slice("ADAE", "AESER", anchored = TRUE),
    teal_slice("ADAE", id = "aerel",
               expr = "AEREL == 'Y' & AETOXGR %in% c('3', '4', '5')",
               title = "Grade 3+ Related Events"),
    teal_slice("ADSL", "SEX", id = "sex", fixed = TRUE),
    teal_slice("ADSL", id = "age",
               expr = "AGE >= 18 & AGE <= 30", title = "Young Adult"),
    module_specific = TRUE,
    mapping = list(
      "Demographic Table" = c("saffl"),
      "AE Table" = c("saffl", "aerel"),
      global_filters = c("sex", "age")
    ),
    count_type = "all"
  )
)

shinyApp(app$ui, app$server)
```

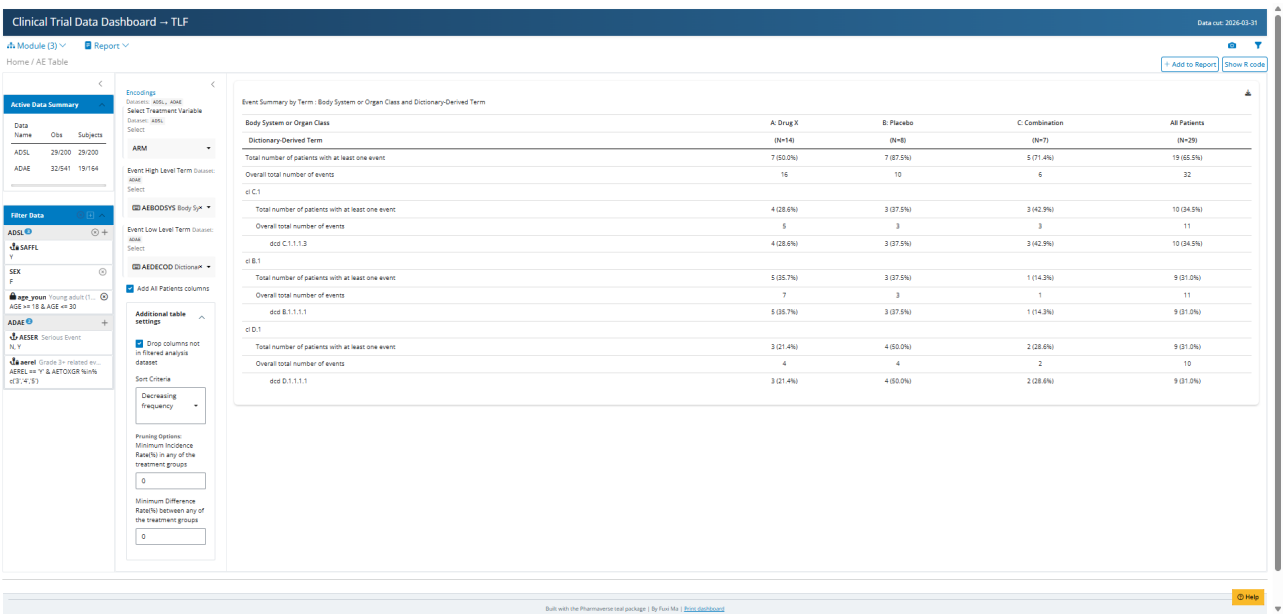
3 Dashboard Snapshots

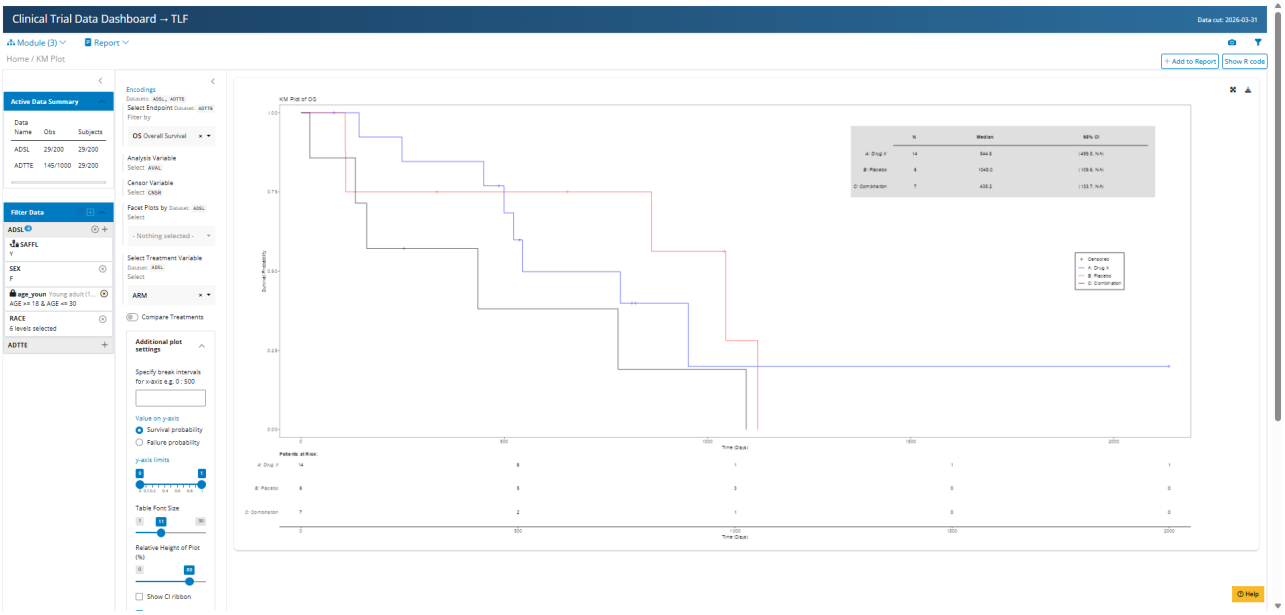
3.1 DM Table



Description of Planned Arm	A: Drug X	B: Placebo	C: Combination	All Patients
	(N=14)	(N=8)	(N=7)	(N=29)
Sex				
n	14	8	7	29
F	14 (100%)	8 (100%)	7 (100%)	29 (100%)
M	0	0	0	0
Race				
n	14	8	7	29
ASIAN	10 (71.4%)	5 (62.5%)	2 (28.6%)	17 (58.6%)
BLACK OR AFRICAN AMERICAN	2 (14.3%)	1 (12.5%)	3 (42.9%)	6 (20.7%)
WHITE	1 (7.1%)	2 (25%)	1 (14.3%)	4 (13.8%)
AMERICAN INDIAN OR ALASKA NATIVE	0	0	1 (14.3%)	1 (3.4%)
MULTIPLE	1 (7.1%)	0	0	1 (3.4%)
NATIVE HAWAIIAN OR OTHER PACIFIC ISLANDER	0	0	0	0
OTHER	0	0	0	0
UNKNOWN	0	0	0	0
Age				
n	14	8	7	29
Mean (SD)	26.6 (1.8)	27.6 (2.6)	27.5 (1.8)	27.1 (2.0)
Mean 95% CI	(25.55, 27.66)	(25.45, 29.80)	(25.80, 29.16)	(26.32, 27.88)
Geometric Mean	26.5	27.5	27.4	27.0
Median	26.7	28.8	27.9	26.9
Median 95% CI	(25.28, 28.24)	(23.34, 29.75)	(25.44, 29.59)	(25.74, 28.35)
25% and 75%-ile	25.5 - 28.2	25.5 - 29.6	25.5 - 29.4	25.5 - 28.6
Min - Max	22.4 - 29.4	23.3 - 29.8	25.4 - 29.6	22.4 - 29.8

3.2 AE Table



Body System or Organ Class	A: Drug X	B: Placebo	C: Combination	All Patients
Dictionary-Derived Term	(N=14)	(N=8)	(N=7)	(N=29)
Total number of patients with at least one event	7 (50.0%)	7 (87.5%)	5 (71.4%)	19 (65.5%)
Overall total number of events	16	10	6	32
cl C.1				
Total number of patients with at least one event	4 (28.6%)	3 (37.5%)	3 (42.9%)	10 (34.5%)
Overall total number of events	5	3	3	11
dcd C.1.1.1.3	4 (28.6%)	3 (37.5%)	3 (42.9%)	10 (34.5%)
cl B.1				
Total number of patients with at least one event	5 (35.7%)	3 (37.5%)	1 (14.3%)	9 (31.0%)
Overall total number of events	7	3	1	11
dcd B.1.1.1.1	5 (35.7%)	3 (37.5%)	1 (14.3%)	9 (31.0%)
cl D.1				
Total number of patients with at least one event	3 (21.4%)	4 (50.0%)	2 (28.6%)	9 (31.0%)
Overall total number of events	4	4	2	10
dcd D.1.1.1.1	3 (21.4%)	4 (50.0%)	2 (28.6%)	9 (31.0%)

3.3 Kaplan-Meier Plot